THE ZEROTH FITTING IDEAL OF THE DIFFERENTIALS CUTS OUT THE RAMIFICATION LOCUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The complement of the zeroth Fitting locus is where the module vanishes. Lemma lemma-on-subscheme-cut-out-by-Fit-0, used at proof lines 1607-1609. Describes the complement of Z.
- Unramified morphisms are those with zero differentials. Morphisms, Lemma morphisms-lemma-unramified-omega-zero, used at proof lines 1609-1611. Identifies that locus with the unramified locus.

Source reference. Original lemma 0CYX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16257–16257. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02G5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3216–3216. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. The closed subscheme $Z \subset X$ cut out by the 0th fitting ideal of $\Omega_{X/S}$ is exactly the set of points where f is not unramified.*

Proof. By Lemma 0CYX the complement of Z is exactly the locus where $\Omega_{X/S}$ is zero. This is exactly the set of points where f is unramified by Morphisms, Lemma 02G5. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.